MILITARY SPECIFICATION

POWER SUPPLY, TRANSISTORIZED, 16-KILOVOLT

1. SCOPE

1.1 This specification covers one type of a transistorized, 16-kilovolt direct current (dc) power supply having a 1.5-volt dc input.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ–N–290 —Nickel Plating (Electrodeposited).
PPP–B–601 —Boxes, Wood, Cleated-Plywood.
PPP–B–621 —Boxes, Wood, Nailed and Lock-Corner.
PPP–B–636 —Box, Fiberboard

MILITARY

MIL–R–26474 —Reliability Requirements for Production Ground Electronic Equipment.

STANDARDS

MILITARY

MIL–STD–105 —Sampling Procedures and Tables for Inspection by Attributes.
MIL–STD–129 —Marking for Shipment and Storage.
MIL–STD–130 —Identification Marking of U.S. Military Property.
MIL–STD–202 —Test Methods for Electronic and Electrical Component Parts.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 **Other publications.** The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules

(Application for copies should be addressed to the Official Classification Committee, 1 Park Avenue at 33d Street, New York, N.Y. 10016.)

AMERICAN TRUCKING ASSOCIATION
National Motor Freight Classification Rules

(Application for copies should be addressed to the American Trucking Association 1616 P Street NW, Washington, D.C. 20036.)

3. REQUIREMENTS

3.1 **Description.** The power supply shall be a transistorized, solid-state, electronic device for converting 1.5-volt dc input to 16.0-kilovolt dc output and shall conform to figures 1 and 2 and as specified herein. The outside case of the power supply shall be of metal suitable for nickel plating.

3.2 **Preproduction model.** As soon as practicable after award of a contract, the supplier shall furnish three power supplies to prove, prior to starting production, that his production methods and choice of design detail will produce power supplies that comply with the requirements of this specification. Examina-

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tion and tests shall be those specified herein. Any changes or deviations from the preproduction model during production shall be subject to the approval of the contracting officer. Approval of the preproduction model by the activity concerned shall not relieve the supplier of his obligation to furnish power supplies conforming to this specification.

3.3 **Material.** Material shall be as specified herein. Materials not specified shall be selected by the supplier and shall be subject to all provisions of this specification.

3.3.1 *Header insert.* The header insert surfaces shall be in the shape of convolutions as shown in figure 2. The material for the header insert shall be polytrifluoromonochloroethylene or other suitable material.

3.4 **Storage and thermal shock.** The power supply shall not be damaged (see 6.4.1) by storage for 4 hours at 155° F. at the highest possible relative humidity or 3 hours at minus 65° F; or by thermal shock from minus 65° F. to room temperature (70° F. plus or minus 5° F.).

3.5 **Shock.** The power supply shall not be damaged when subjected to shock impacts of 100 g's (see 6.4.2) applied six times parallel with and six times perpendicular to the longitudinal axis of the power supply.

3.6 **Vibration.** The power supply shall not be damaged when subjected to a simple harmonic motion in accordance with Standard MIL–STD–202, test method 201A.

3.7 **Rise time.** The power supply shall deliver 15.0 kilovolts when connected to a load of 5×10^{-7} amperes within 5 seconds after energizing the input circuit.

3.8 **Short circuit.** The power supply shall not be damaged when a direct short circuit is applied for at least 1 minute between the output terminal and the case ground while operating nominally (6.4.3).

3.9 **Reliability.** The power supply shall exhibit a mean time of 1,000 hours between failures in accordance with Specification MIL–R–26474.

3.10 **Electrical characteristics.** The power supply shall have the input terminal positive and output terminal negative with respect to the metallic case and the maximum peak-to-peak ripple voltage superimposed on the dc output voltage limited to 1.5 percent of the dc output voltage. The power supply shall be capable of operating under the conditions specified in table I with an input voltage of 1.50 plus or minus 0.02 volt dc.

TABLE I. *Operating conditions*

F. temp	Relative humidity	Input	Load current	Output, excluding ripple
	Percent	*Milliamperes*	*Amperes*	*Kilovolts*
125°	95–99	24	1×10^{-9}	15.8–16.0
		26	1×10^{-7}	15.5–16.0
		30	5×10^{-7}	15.0–16.0
		35	1×10^{-6}	14.0–16.0
70°±5°	90–95	24	1×10^{-9}	15.8–16.0
		26	1×10^{-7}	15.5–16.0
		30	5×10^{-7}	15.0–16.0
		35	1×10^{-6}	14.0–16.0
−40°		24	1×10^{-9}	15.8–16.0
		26	1×10^{-7}	15.5–16.0
		30	5×10^{-7}	15.0–16.0
		35	1×10^{-6}	14.0–16.0

3.11 **Identification marking.** The power supply shall be identified in accordance with *Standard MIL–STD–130.*

3.12 **Finish.** All exterior metallic surfaces of the power supply, including the high voltage pin and the low-voltage end shall be copper flashed and nickel plated to a minimum thickness of 0.0002 inch in accordance with Specification QQ–N–290, class 1.

3.13 **Workmanship.** The power supply shall be free from defects that would impair performance or serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 **Responsibility for inspection.** Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as other-

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wise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 **Classification of inspection.** Inspection shall be classified as follows:

(a) Preproduction inspection of the power supplies.
(b) Quality conformance inspection of the power supplies.
(c) Inspection of preparation for delivery.

4.3 **Preproduction inspection.**

4.3.1 *Examination.* Each power supply shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 *Tests.* Each power supply shall be tested as specified in 4.5.2. Failure of any test shall be cause for rejection.

4.4 **Quality conformance inspection.**

4.4.1 *Sampling.* Sampling for examination and tests shall be in accordance with Standard MIL–STD–105. Inspection level S–2 shall be used for lot sizes 2 to 25; inspection level S–4 shall be used for lot sizes 26 or larger.

4.4.2 *Examination.* Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. AQL shall be 10 percent defective.

4.4.3 *Tests.*

4.4.3.1 *Individual.* Each power supply shall be tested as specified in 4.5.2.4.2.

4.4.3.2 *Sample.* Samples selected in accordance with 4.4.1 shall be tested as specified in 4.5.2.1, 4.5.2.2, 4.5.2.3, 4.5.2.4.3, 4.5.2.5, and 4.5.2.6. AQL shall be 10 percent defective.

4.5 **Inspection procedure.**

4.5.1 *Examination.* The power supply shall be examined for the following defects:

101. Dimensions not as specified.
102. Plating not as specified.
103. Identification marking not as specified.
104. Material not as specified.
105. Workmanship not as specified.

4.5.2 *Tests.*

4.5.2.1 *Vibration.* The power supply shall be tested in accordance with Standard MIL–STD–202, test method 201A. The power supply shall be nominally operated prior to the vibration test. The power supply shall be strapped to a suitable test fixture that is free from resonances over the test range and vibrated for a period of 30 minutes in each of the three mutually perpendicular directions. After vibration, the power supply shall again be nominally operated. Evidence of damage (see 6.4.1) or failure to produce the specified current and voltage values shall constitute failure of this test.

4.5.2.2 *Shock.* The power supply shall be tested in accordance with Standard MIL–STD–202, test method 202A. The power supply shall be fastened to the carriage assembly by being strapped to a suitable plate which will assure rigid fastening, raised to a height necessary for an acceleration of 100 g's, and dropped six times parallel with and six times perpendicular to the longitudinal axis of the power supply. After the drops, the power supply shall be nominally operated. Evidence of damage or failure to produce the specified voltage and current values shall constitute failure of this test.

4.5.2.3 *Short circuit.* The power supply shall be energized and the output terminals shall be shorted together for 1 minute. The short shall be removed and the power supply shall be nominally operated. Evidence of damage or failure to produce the specified voltage and current values shall constitute failure of this test.

4.5.2.4 *Temperature.*

4.5.2.4.1 *Storage.* The power supply shall be placed in a humidity-temperature controlled chamber and the temperature shall be raised to 155° F. with the highest possible

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relative humidity and held for 4 hours. The chamber temperature shall then be lowered to room temperature and the power supply shall be nominally operated and examined. The chamber temperature shall then be lowered to minus 65° F. After 3 hours, the power supply shall be removed to room temperature. After 1 hour, the power supply shall be nominally operated. The above cycle from room temperature to minus 65° F. to room temperature shall be performed four more times. Evidence of damage or failure to produce the specified voltage and current values shall constitute failure of this test.

4.5.2.4.2 *Electrical characteristics.* The power supply shall be connected to a power source of 1.50 plus or minus 0.02 volt with the output connected to a variable load as shown in figure 3 and operated at the temperature, relative humidity, and loads specified in table I. The power supply shall be soaked for a period of 1 hour at each temperature before the power is turned on and the load applied. Inability to obtain the voltage and current specified in 3.10 shall constitute failure of this test.

4.5.2.4.3 *Rise time.* The rise time of the power supply shall be determined at 125° F., 70° plus or minus 5° F., and minus 40° F. under a load of 5×10^{-7} amperes and 15 kilovolts. The power supply shall be soaked for 1 hour at each temperature before the power is turned on. Inability of the power supply to reach 15 kilovolts at 5×10^{-7} ampere load in 5 seconds shall constitute failure of this test.

4.5.2.5 *Ripple voltage.* The power supply ripple voltage shall be determined using the circuit shown in figure 4. Inability of the power supply to maintain a peak-to-peak ripple voltage under 1.5 percent of the dc output voltage shall constitute failure of this test.

4.5.2.6 *Reliability.* The power supply (with an input voltage of 1.50 plus or minus 0.02) shall be connected to a load of 5×10^{-7} amperes at 15.0 to 16.0 kilovolts and operated in accordance with Specification MIL–R–26474. The total operating hours completed previous to this test may be considered as part of this test. Inability to perform as specified shall constitute failure of this test.

4.6 Inspection of preparation for delivery. The packaging, packing, and marking shall be examined and tested to determine compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging.

5.1.1 *Levels A and C.* Each power supply shall be cushioned and packaged in a close-fitting paperboard carton.

5.1.2 *Intermediate packaging.* The power supplies, packaged as specified in 5.1.1, shall be intermediate packaged in quantities of 25 in boxes conforming to Specification PPP–B–636, class 2, W5c, style optional. After closure, the boxes shall be sealed with tape in accordance with the appendix to the box specifications as specified for interior containers.

5.2 Packing. Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 *Level A.* The power supplies shall be packed in boxes conforming to Specification PPP–B–621, class 2, style optional, or Specification PPP–B–601, oversea type, style optional. The boxes shall be strapped in accordance with the appendix to the applicable box specification. Strapping shall be zinc coated. The gross weight of each box shall not exceed 200 pounds.

5.2.2 *Level B.* The power supplies shall be packed as specified for level A except the containers shall be domestic type, style, and class. The boxes shall be strapped in accordance with the appendix to the applicable box specification.

5.2.3 *Level C.* The power supplies shall be packed to assure undamaged delivery from the supplier to the initial destination in containers complying with Uniform Freight Classification Rules, National Motor Freight Classification Rules, or other carrier rules applicable to the mode of transportation.

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5.3 **Marking.** Packages and shipping containers shall be marked in accordance with MIL–STD–129.

6. NOTES

6.1 **Intended use.** The power supply is intended to be used to supply the high voltage for the type 6914 infrared image converter tube.

6.2 **Ordering data.** Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Level of packing required (see 5.2).

6.3 **Description of test equipment.**

6.3.1 *Variable load.* The variable load shown in figures 3 and 4 consists of an image tube type 6914 and a light source with variable voltage control, both mounted within a lightproof box

6.4 **Definitions.**

6.4.1 *Damage.* Damage is breakage, compound leakage, or corrosion of any finish, or deterioration of dielectric characteristics.

6.4.2 *g.*—g is the force which results in an acceleration or deceleration of 32.17 feet per second per second.

6.4.3 *Nominal operation.* Nominal operation of the power supply requires an output current of 5×10^{-7} amperes at an output voltage between 15.0 and 16.0 kilovolts using an input voltage of 1.50 plus or minus 0.02 volt and operating for a period of 10 minutes.

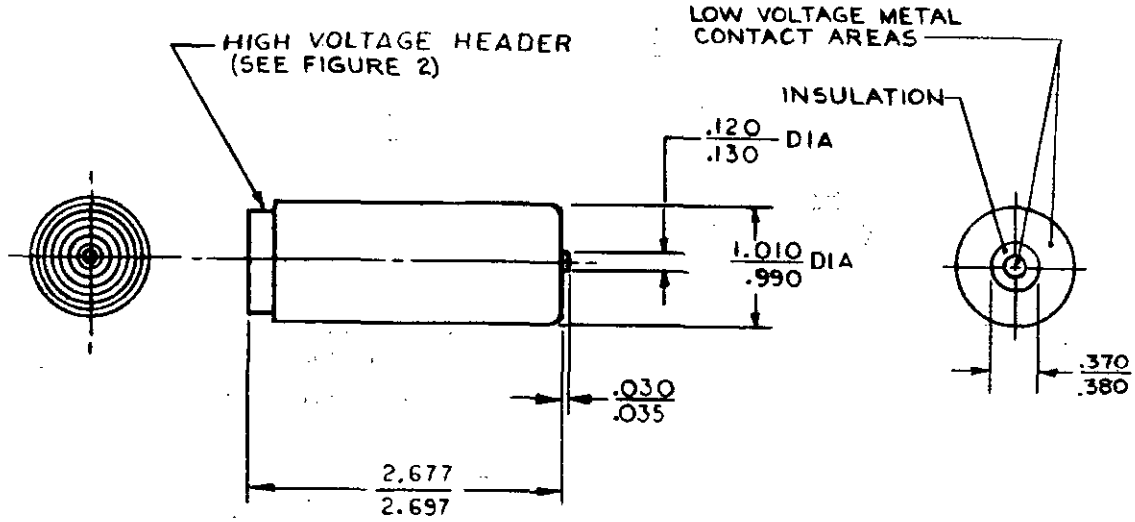
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NOTE:
DIMENSIONS
ARE IN INCHES.
SCALE: 1/1

FIGURE 1. *16KV power supply assembly.*

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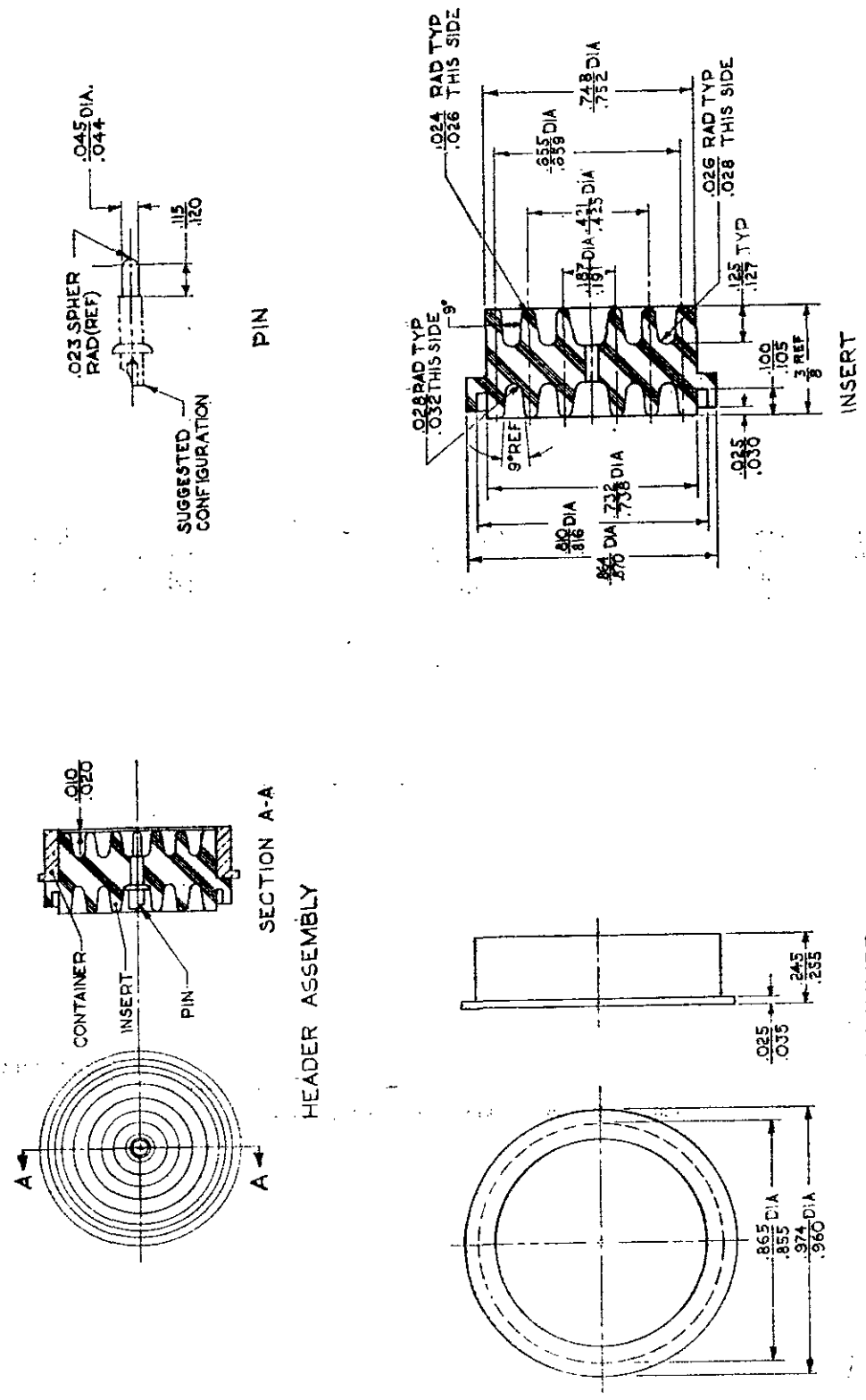


FIGURE 2. *High voltage header assembly.*

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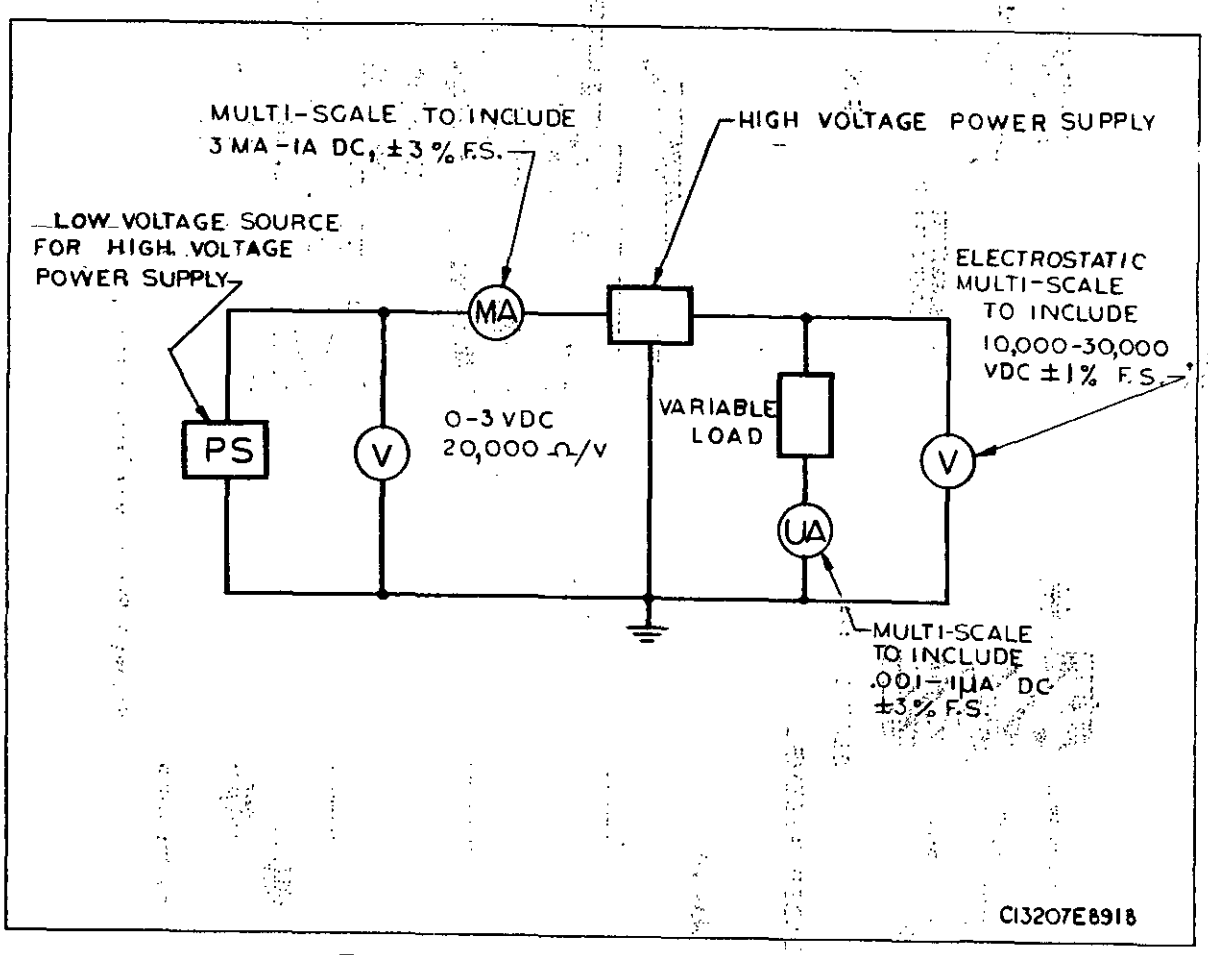


FIGURE 3. *Electrical characteristics test circuit.*

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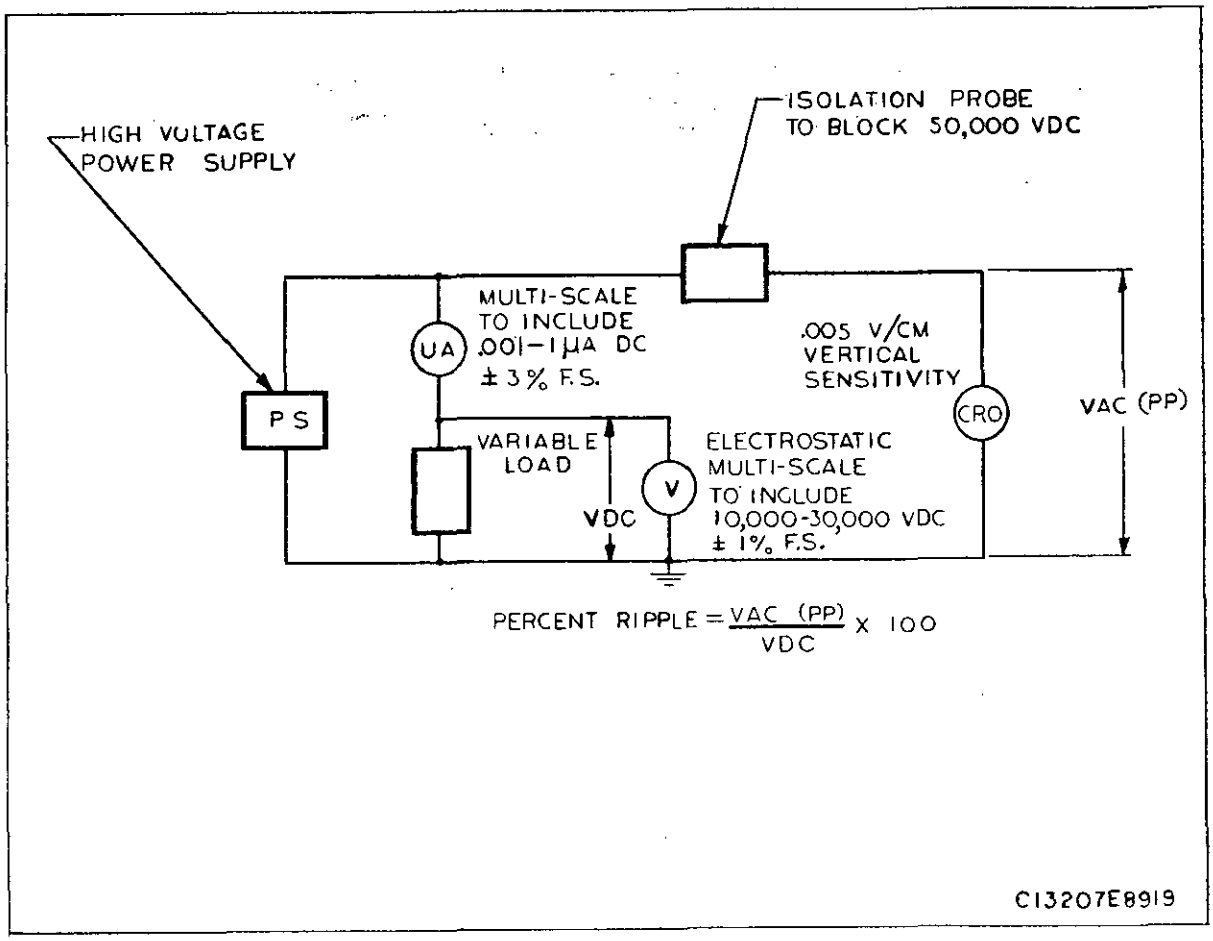


FIGURE 4. *Ripple test circuit.*

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